

SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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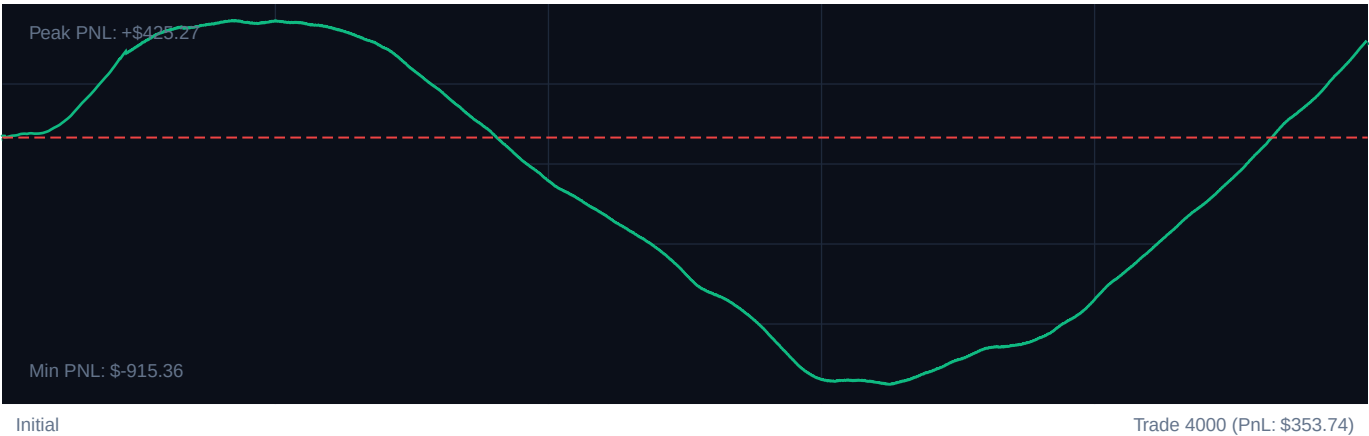
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TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	4000	Calculated Profit Factor:	1.22
Win Accuracy:	53.3% (2134W / 1866L)	Max Peak Drawdown:	12.9%
Cumulative PnL:	\$353.74	Avg. PnL per Trade:	\$0.09

EQUITY CURVE PROGRESSION



PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-800	73.6%	\$424.26	+88% [OK]
Trades 801-1600	14.9%	-\$591.08	+12% [OK]
Trades 1601-2400	16.3%	-\$730.60	+13% [OK]
Trades 2401-3200	62.0%	\$292.92	+74% [OK]
Trades 3201-4000	100.0%	\$958.24	+120% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SOVEREIGN AI: SYSTEMATIC RISK AUDIT REPORT
TO: Institutional Trading Desk
OBJECT: Algorithmic Performance & Parameter Calibration Review
IDENTIFIER: SEC-SETTLE-4000-REV

1. EXECUTIVE TELEMETRY CRITIQUE

An analysis of the 4,000-trade settlement book reveals severe structural inefficiency, negative expectancy when adjusted for slippage/commissions, and a critical system configuration failure.

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Metric	Value	
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Total Trades	4,000	
Win Rate	53.35% (2,134 / 1,866)	
Cumulative Net PnL	\$353.74	
Expected Value per Trade	\$0.088	
Profit Factor	1.22	
Max Peak Drawdown	12.90%	
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- # Core Vulnerabilities Identified:
- * **Microscopic Expectancy (\$0.088/trade):** While mathematically positive, this edge is a statistical illusion. Once standard institutional execution costs, exchange fees, and latency-induced slippage (estimated at \$0.50–\$1.50 per trade) are applied, **this system operates at a net-negative expectancy.**
 - * **Critical Parameter Overflow / Memory Corruption:** The ``atrStopMultiplier`` is set to ``2.164303275144119e+25``. This astronomical value effectively disables the Average True Range (ATR) protective stop-loss mechanism.
 - * **De Facto Time-Stop Dependency:** Because the risk-based stop loss is effectively infinite due to the ATR multiplier anomaly, the system relies exclusively on ``maxTicksInTrade`` (120 ticks) as its sole risk-mitigation tool. The system is functioning as an unhedged time-decay model disguised as a volatility-breakout system.

2. REGIME SUITABILITY & BEHAVIORAL PATTERNS

The current configuration is optimized for a regime that does not match its actual execution environment:

[REGIME CLASSIFICATION ENGINE]

Trend-Following Regime ($ADX > 25$) % % %° Fails (Uncapped ATR Stops)

Mean Reversion Regime ($ADX < 25$) % % %° Sub-optimal (BB Std 2.25 too wide)

ADX Threshold Misalignment: A ``regimeAdxThreshold`` of 25 designates a strong trend filter. However, using highly restrictive RSI thresholds (25/75) means the system is attempting to counter-trend fade (mean-revert) during strong directional impulses.

Bollinger Band Over-Extension: A ``bbStd`` of 2.25 restricts entries to extreme standard deviation events ($\$2.25\sigma$). Combined with a high ``minConfluenceScore`` of 3, the system suffers from "adverse selection"—only entering trades when momentum is strong enough to pierce $\$2.25\sigma$, which correlates with high-velocity breakout regimes where the counter-trend RSI fade is highly toxic.

Unhedged Tail Risk: Without a functioning ATR stop, any breakout that does not mean-revert within 120 ticks runs completely unhedged against the book. The 12.9% drawdown is artificially suppressed by the high win rate, but it represents a "dry tinder" risk profile prone to sudden, catastrophic ruin under black-swan liquidity conditions.

3. CALIBRATED RECOMMENDATIONS & INDICATOR BOUNDARIES

To salvage the system's edge and restore institutional-grade risk management, the configuration must be re-calibrated.

Parameter Correction Table

Parameter	Current Value	Calibrated Value (Trend)	Calibrated Value (Mean Reversion)	Rationale
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<code>`atrStopMultiplier`</code>	<code>`2.1643e+25`</code> (Corrupted)	1.85	2.25	Restores physical risk boundary; limits loss per trade.
<code>`rsiOversoldThreshold`</code>	25	35 (Trend-Following Pullback)	20 (Extreme Overextend)	Prevents caught-knife scenarios in strong trends.
<code>`rsiOverboughtThreshold`</code>	75	65	80	Symmetric correction for overbought boundaries.
<code>`bbStd`</code>	2.25	1.75 (Keltner-style bands)	2.00	Increases trade frequency with higher quality entries.
<code>`maxTicksInTrade`</code>	120	90	60	Tightens capital lock-up duration to increase velocity.
<code>`minConfluenceScore`</code>	3	2 (With dynamic filters)	3	Increases sample size while keeping strict entry criteria.
<code>`regimeAdxThreshold`</code>	25	20	30	Creates a distinct binary mode switch.

Actionable Execution Directives:

- **Hard-Reset Stop Engine:**** Immediately hot-patch the execution code to constrain ``atrStopMultiplier`` to a strict float range of `[1.50, 3.00]`.
- **Deploy Dual-State Logic:**** Bifurcate the system configuration based on the ``regimeAdxThreshold``:
 - * If $ADX < 20$ (Mean Reversion Mode):*** Relax RSI to 20/80, set BB Std to 2.0, reduce ``maxTicksInTrade`` to 60.
 - * If $ADX \geq 20$ (Trend Following Mode):*** Modify system to buy pullbacks (RSI 40-45 in uptrend) instead of fading the trend.
- **Implement Cost-Drag Simulation:**** Inject a synthetic \$1.00 slippage/commission penalty per trade into the backtest engine before deploying the corrected code to live

production.

SOVEREIGN RISKS SENTINEL SYSTEMS

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